Fiscal Year:	FY 2013	**Task Last Updated:**	FY 03/21/2013
PI Name:	Yu, Nan Ph.D.		
Project Title:	JPL Participation in ESA ACES (Atomic Clock Ensemble in Space) Worldwide Clock Comparison Campaign		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Fundamental Physics Rapid Turn Around Flight Opportunity
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Burt, Eric (Jet Propulsion Laboratory) Prestage, John (Jet Propulsion Laboratory) Tjoelker, Robert (Jet Propulsion Laboratory)		
Grant/Contract No.:	JPL Task Plan number 71-15743		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Tremendous progress has been made recently in the U.S. and other countries in advancing atomic clock technology and its applications for the benefit of science, national interests, and society in general. Space provides a unique environment in which many benefits can be realized. The European Space Agency (ESA) is readying the ACES (Atomic Clock Ensemble in Space) payload for the International Space Station (ISS) for an anticipated 2013 launch. Some of the key objectives of ACES will be achieved by global time transfer and clock comparisons with the most advanced clocks at multiple worldwide sites. There is an opportunity for investigators from the leading US clock and metrology labs to participate in the clock comparison campaign, enhancing the overall science objective of ACES and at the same time demonstrating and providing science benefit from advanced US clock technology. JPL is one of the major US time and frequency metrology institutions serving NASA missions and Deep Space Network needs. It was the NASA center for the microgravity LCAP program (Laser Cooling and Atomic Physics) in charge of developing the cold atom clock payload for PARCS (Primary Atomic Reference Clock in Space). JPL has also developed a unique atomic frequency standard that has demonstrated an exceptional long-term stability. We propose to collaborate with ESA on the ACES mission by participating in the ACES global clock campaign with JPL's ultra-stable mercury ion clock. During a proposed six-year performance period, we will: 1) contribute significantly to the ACES mission science objectives by providing unique clock references from a Western US site; 2) Provide and demonstrate a high-impact application for JPL's high performance time keeping clock technology; and 3) re-establish NASA's involvement in ISS Clock Science and initiate a healthy collaboration between NASA and ESA in the area of clock and quantum sensors. The task is to collaborate with ESA on the ACES mission by participating in the ACES global clock comparison campaign and provide frequency and time references derived from JPL's ultra-stable mercury ion clock. These references will be connected to a microwave link ground terminal provided by ACES and installed at JPL. Clock comparisons will be carried out over the microwave link with ESA clocks on the ISS. As JPL is only ground comparison site located in the western USA and the mercury ion clock is the only one of its kind in the global clock comparison campaign, JPL's participation will contribute significantly to the overall ACES mission science objectives, while simultaneously providing a high-impact application for JPL's high performance time keeping clock technology. The task will also help re-establish NASA's involvement in ISS clock science and enable future collaboration between NASA and ESA in the area of clocks and quantum sensors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Advanced atomic clocks are critical to the interests of NASA and the nation. Telecommunication and navigation (e.g., GPS) are well-known applications of space clocks. Within NASA, high performance atomic clocks and frequency and timing capabilities enable operation of the NASA Deep Space Network and provide fundamental tools for precise measurements exploring the laws of Nature. Progress in advanced clock technology development, both on ground and for space, is rapidly being made in many countries. Participation by the U.S. in the ACES mission contributes to the science objectives of the ISS consistent with the 2011 NRC report recommendations in Fundamental physics and demonstrates the capability and potential of NASA mercury ion clock technology.
Task Progress:	The task is to collaborate with ESA on the ACES mission by participating in the ACES global clock comparison campaign and to provide frequency and time references derived from JPL's ultra-stable mercury ion clock. These references will be connected to a microwave link ground terminal (GT) provided and calibrated by ACES and installed at JPL. Clock comparisons will be carried out over the microwave link with ESA clocks on the ISS. As JPL is located in the western USA and the mercury ion clock is one of its kind in the global clock comparison network, JPL's participation will contribute significantly to the overall ACES mission science objectives, while simultaneously providing a high-impact application for JPL's high performance time keeping clock technology. The task will also help re-establish NASA's involvement in ISS clock science and enable future collaboration between NASA and ESA in the area of clocks and quantum sensors. In the last year, we have carried out the baseline ground clock upgrade effort while leveraging a concurrent new generation clock development. We have also identified the suitable site at JPL for the installation of the microwave link ground terminal and are in the process of its engineering design. The construction will begin next FY. Finally, we have initiated science study for the planned global clock comparison, contacted and discussed with ESA colleagues for the ACES task coordination and future opportunities. We have presented our status at a number of US and international conferences.
Bibliography Type:	Description: (Last Updated: 10/28/2020)
Abstracts for Journals and Proceedings	Yu N. "Fundamental Physics Measurements with Precision Clocks Aboard ISS and on the Ground." 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. 28th Annual Meeting of the American Society for Gravitational and Space Research, New Orleans, LA, November 27-December 2, 2012. Program and abstracts, p. 66. https://www.asgsr.org/images/stories/pdf/2012_AnnualMeeting_Program.pdf , Nov-2012
Abstracts for Journals and Proceedings	Yu N. "Atomic Clocks and Precision Measurements on ISS." 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. Presentation by Robert Thompson, in PI Yu's lab at JPL. 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. , Jun-2012